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Age of linear enamel hypoplasia formation based on Massler and colleagues' and Reid and Dean's standards in a Polish sample dated to 13th–18th century CE



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ABSTRACT

In the study of enamel hypoplasia formation ages basically two standards of dental development are being applied. According to some authors, different methods produce different results, thus it is critical to decide which method to use. This study focuses on the comparison of three methods for estimating ages of linear enamel hypoplasia (LEH) formation.

The skeletal material derives from three burial grounds localized on the territory of Poland, dated to 13th–18th c. CE. In total 642 teeth of 77 individuals were qualified for the study. The position of LEHs on the crown surface was measured and then converted to age at formation using the regression equations proposed by Goodman and Rose, Goodman and Song, and the data provided by Reid and Dean.

There were 51.9% of individuals and 17.9% of teeth affected by LEH. The lowest age estimates were provided by the Goodman and Rose's method then higher by the Goodman and Song's method and the highest using the Reid and Dean's data. The age ranges and peaks of LEH provided by the three methods differed more for the maxillary than for the mandibular teeth.

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Crown formation ages supplied by Reid and Dean account for intra- and inter-tooth as well as inter-population variation in the timing of tooth development and also for variation in crown heights. This approach undoubtedly increases reliability of the results. Thus, it seems advisable to use the Reid and Dean's standard when calculating ages of enamel hypoplasia formation.

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Introduction

In skeletal studies of human populations teeth show considerable potential, since they can provide much information about age at death, diet, diseases, health status, genetic affiliation, as well as socio-economic status, migrations and behavior of past people (Hillson, 2005; Kelley and Larsen, 1991; Ogden, 2008). Data obtained from dental research are considered reliable, since teeth are the most highly mineralized body structures resistant to taphonomic changes; there are sometimes the only preserved elements of the human skeleton (Ogden, 2008). Tooth enamel is formed in a rhythmic and regular manners and is sensitive to nutritional and disease factors (Goodman et al., 1987; Infante and Gillespie, 1974; Sarnat and Schour, 1942). It does not undergo remodeling once formed and therefore retains the episodes of developmental disturbances which have occurred during the individual's infancy and childhood (Buikstra and Ubelaker, 1994; Goodman and Rose, 1990). Stress episodes cause deficiencies in enamel thickness, called enamel hypoplasia. Researchers usually focus on linear forms of dental hypoplasia (LEH), as they reflect the chronological development of tooth enamel (Cucina and İşcan, 1997; Goodman and Song, 1999; King et al., 2005; Ritzman et al., 2008). The metabolic insult affects only this part of the tooth which is just being formed, thus the LEH's location on the crown surface can be converted into the age of its occurrence (Buikstra and Ubelaker, 1994; Larsen, 1987). These specific features of dental enamel, together with the possibility of chronological reconstruction of developmental defects have made LEH most widely used stress marker in skeletal studies (Goodman and Rose, 1990).

The majority of researchers consider higher prevalence of LEH as evidence of poor living conditions and exposure to elevated level of stress in childhood (e.g. Goodman et al., 1980; Lanphear, 1990). Some researchers have raised questions related to the study of health status using skeletal samples by describing the phenomenon called an "osteological paradox" (Wood et al., 1992). This idea caused long-term discussion among the skeletal biologists (Byers, 1994; Goodman, 1993; Jackes, 1993; Milner et al., 2000; Saunders and Hoppa, 1993; Wood and Milner, 1994; Wright and Yoder, 2003). However, the results of numerous studies enable us to contend that the potential error related to the interpretations based on the skeletal remains is not as large as Wood et al. (1992) thought to be (Cohen, 1997; Larsen, 1987; Saunders and Hoppa, 1993, after Ortner, 2003). Some studies compared LEH prevalence in advantaged and disadvantaged samples, and showed that in both, skeletal samples (Bennike et al., 2005) and in modern populations (Aine et al., 2000) more defects were found in the groups suffering from compromised health. Moreover, some authors claim that the errors related to methodology, such as those deriving from the use of different tooth developmental standards, are more significant than previously thought (Hoppa and FitzGerald, 1999).

Several methods for LEH timing estimation have been developed, based on measurements of hypoplastic defects from the cemento-enamel junction to the midline of LEH which were then converted to age at formation using a standard of enamel crown development. Mainly, two developmental standards are being used, the one originally created by Massler et al. (1941) and the other developed by Reid and Dean (2000, 2006). However, together with the development of new techniques of enamel defect research (i.e. scanning electron microscopy), some reservations were raised against the accuracy of the former (Goodman and Song, 1999; Reid and Dean, 2006), as it does not take into account, among others, the nonlinearity of tooth development, hidden cuspal enamel, and intra- and interpopulation variation in the timing of tooth growth and variation in the crown heights. The studies of Goodman and Song (1999) and Krenz-Niedbala and Kozłowski (2013) showed that the method correcting the

Massler et al.'s data only for hidden enamel shifted the ages of LEH formation to later categories, as compared with the results based on uncorrected Massler et al.'s standard.

Histological studies carried out by Reid and Dean (2006) showed that the enamel defects formed prior to the first year of age are practically invisible on the crown surface. They proposed a model for LEH timing assessment accounting for variation in tooth initiation times, the times taken to form various fractional stages of the crown, differences in the development of different tooth types, and related inter-population variation. Ritzman et al. (2008) and Martin et al. (2008) were the first to compare ages of LEH formation estimated using Massler et al.'s standards (chart and regression-based methods) with ages produced using Reid and Dean's crown formation times. Ritzman et al. (2008) found that the latter are significantly higher than the former, and particularly for earlier forming defects, while Martin et al. (2008) found that the average ages at LEH formation from the decile chart are higher for some tooth types and lower for other, with the differences ranging from below 1 month to 4 months. In recent studies data on tooth crown formation times provided by Reid and Dean (2000, 2006) have been increasingly used. However, still most LEH data, which have been published to date, were based on the tooth mineralization diagram established by Massler et al. (1941), and modified for metric analysis by Swärdstedt (1966); for example, these are studies by Moggi-Cecchi et al. (1994), Palubeckaitė et al. (2002), Obertová (2005), Staniowski et al. (2008). Different results may produce different interpretation. Undoubtedly, imprecise determination of hypoplasia timing can lead to faulty conclusions. Thus, it seems critical to investigate how much the results obtained with different methods differ from each other.

The aim of the present study is to find out the magnitude of differences in LEH timing estimation introduced by the methods based on the Massler et al.'s and Reid and Dean's data in a Polish late medieval/early historical sample.

Material

The skeletal material used in this study comes from three late medieval/early historical burial grounds within the territory of Poland. The first cemetery is called Łekno Ł3 site, which was used from the 13th to the 16th centuries, the second one is Łekno Ł19 site, dated to the period between the 16th to the 18th centuries, and the third site is Słaboszewo, that functioned between the second half of the 14th and the first half of the 17th century (Fig. 1).

The samples from Łekno and Słaboszewo were studied jointly, given their close proximity (c. 60 km) and similar chronology. Out of 315 adult individuals only 77 could have been included in the analysis of enamel hypoplasia. The dentition of the remaining adults did not meet the criteria for the study. The chosen individuals had the teeth of at least two categories (including at least one anterior tooth), only slightly worn and with visible enamel (no dental calculus).

The material studied for LEH consists of considerably more male than female dentitions (Table 1). This is caused by unequal sex proportion in the sample from Łekno, where the female skulls make up less than 15% of the sample. A much smaller number of female than male remains in this group may possibly result from the presence of a Cistercian monastery in Łekno, which may be the reason why more males than females were buried at the cemetery (Wyrwa, 2003).

In the examined sample the most numerous are young adults, 20–35 years (59.7%), then middle adults, 35–50 years (33.8%), and old adults, 50+ years (6.5%). In total, 642 teeth were studied, including 328 maxillary and 314 mandibular teeth. Among them were 60 central incisors, 96 lateral incisors, 141 canines, 178 first premolars and 167 second premolars.

Table 1
Number of individuals included in the LEH study.

Population	Female		Male		Total	
	N	%	N	%	N	%
Słaboszewo	23	54.8	19	45.2	42	100
Łekno	5	14.3	30	85.7	35	100
Total	28	36.4	49	63.6	77	100



Fig. 1. Map of Poland with the location of the examined sites.

Methods

Age and sex of the examined individuals were assessed using standard methods. Age-at-death was estimated through assessment of cranial suture closure and tooth wear. Sex was determined based on the morphology of the skulls (Buikstra and Ubelaker, 1994; White and Folkens, 2005). Anthropological documentation compiled by E. Miłosz (unpublished), based on the morphology of the entire skeleton, was also used.

Enamel hypoplasia was macroscopically observed and scored on incisors, canines and premolars, according to the recommendations of Goodman and Rose (1990), and Buikstra and Ubelaker (1994). Teeth with severe attrition were excluded from the analysis. Defects were scored if visible to the naked eye and their presence was tested according to whether the enamel depression could be felt with a fingernail (Steckel et al., 2006). All forms of the condition, including pits, were recorded, but for the purpose of this study only linear defects were analyzed. The individuals were scored only if it was possible to observe at least two different tooth types, including obligatorily one anterior tooth. If both antimeres were present, only those LEHs were recorded that matched between the right and left teeth, to ensure they reflect systemic rather than localized stress.

The position of LEH on the crown surface was measured on the labial/buccal surface in the midline from the cemento-enamel junction to the upper (mandible) or lower (maxilla) limit of the defect with a digital caliper calibrated to the nearest 0.01 mm. This approach allows for a calculation of the age of the earliest stressful event. The LEH measurements were then converted to the age at formation using three methods: regression equations proposed by Goodman and Rose (1990), the equations given by Goodman and Song (1999), and the method based on the standard proposed by Reid and Dean (2000, 2006). In the first method, called a conventional method, the chart of enamel development for the permanent teeth, constructed by Massler et al. (1941), then modified by Swardstedt (1966) was used. In this method the authors have assumed a constant rate of enamel formation and have taken into account the age of an individual at the beginning and the end of enamel mineralization for particular tooth types and also the total crown heights. The basic form of the regression equation is as follows:

$$\text{Age} = - \left[\frac{1}{\text{velocity}} \times \text{distance of the LEH in mm from CEJ} \right] + \text{age at crown completion}$$

Table 2

LEH measurements (in mm) and age at LEH formation (in years) for each decile of the crown heights – maxilla [calculated using tooth crown formation times from [Table 4](#) in [Reid and Dean \(2006, p. 333\)](#)].

Crown height (%)	I1		I2		C	
	LEH measurement	Age	LEH measurement	Age	LEH measurement	Age
0–10	12.28–13.63	1.1–1.3	10.46–11.60	1.8–2.0	10.66–11.83	1.7–1.9
11–20	10.91–12.27	1.4–1.6	9.29–10.45	2.1–2.2	9.47–10.65	2.0–2.2
21–30	9.55–10.90	1.7–1.8	8.13–9.28	2.3–2.4	8.29–9.46	2.3–2.4
31–40	8.19–9.54	1.9–2.0	6.97–8.12	2.5–2.7	7.11–8.28	2.5–2.7
41–50	6.83–8.18	2.1–2.4	5.81–6.96	2.8–2.9	5.92–7.10	2.8–3.0
51–60	5.46–6.82	2.5–2.9	4.65–5.80	3.0–3.3	4.74–5.92	3.1–3.4
61–70	4.10–5.45	3.0–3.4	3.49–4.64	3.4–3.7	3.56–4.73	3.5–3.8
71–80	2.74–4.09	3.5–3.9	2.33–3.48	3.8–4.1	2.38–3.55	3.9–4.3
81–90	1.37–2.73	4.0–4.4	1.17–2.32	4.2–4.6	1.19–2.37	4.4–4.8
91–100	0–1.36	4.5–5.0	0–1.16	4.7–5.1	0–1.18	4.9–5.3

In the above equation velocity is given in mm/year, thus it refers to the crown height and the total time of tooth development. The age at defect formation is expressed in years and reflects the developmental age ([Goodman and Song, 1999](#)). The regression equations proposed by [Goodman and Rose \(1990\)](#) have been used for the teeth, that were included in this study ([Table 3](#)).

The second method for the calculation of the individual's age at the time of defect formation was based on the regression equations given by [Goodman and Song \(1999\)](#) and was called a corrected method. These equations were also constructed based on the chart of enamel development for the permanent teeth by [Massler et al. \(1941\)](#), with [Swärdstedt's modification \(1966\)](#). However, they included the time that passes between the initiation of enamel mineralization and the first appearance of enamel on the crown surface, so they account for the hidden cuspal enamel. [Goodman and Song \(1999\)](#) consider buried enamel to be the main source of variation in LEH timing estimation and recommend correction of the timing as directional and biologically justified. By applying corrected and uncorrected Massler's data, we have had the opportunity to discover the magnitude of the effect of buried enamel on the results.

This adjustment considerably increases age at formation of early-developing defects in comparison with the conventional method ([Goodman and Song, 1999](#)). The authors suppose that the most accurate corrections should be included between the proposed in the literature values of 6 months and 12 months, since the first value appears to be an underestimate and the second one an overestimate ([Hillson and Bond, 1997](#)). [Goodman and Song \(1999\)](#) showed the results of the correction for hidden cuspal enamel based on the rounded approximation of the averages of appositional formation times of particular tooth types given by various authors. In this study the same approach was adopted, based on the equations presented by [Goodman and Song in Table 9.4 \(1999, p. 225\)](#).

The third method applied in this study was the one based on the mineralization standard proposed by [Reid and Dean \(2000, 2006\)](#) and called here a decile method. It was developed with the use of histological research of enamel formation times of incisors, canines and molars. The authors divided each tooth into ten equal units (deciles) and calculated the ages of each decile's formation, and also measured the total crown heights. In their studies they accounted for interpopulation variation in the timing of tooth development and also variation in crown heights, since they examined European, American and African dentitions separately. The method assumes different times of enamel initiation, formation of particular deciles, and development of different tooth types as well as related differences that occur between human populations.

In the present study data on the time of enamel formation and mean crown heights for a Northern European population given by [Reid and Dean \(2006\)](#) were used to construct the tables showing the ranges of LEH measurements and ages at defect formation for each decile of maxillary and mandibular tooth ([Tables 2 and 3](#)). This way of data presentation facilitates the practical use of the Reid and Dean's standards. We have used Northern European crown heights instead of mean crown heights from the samples examined in this study, because there weren't enough unworn teeth available (see, for example, [Cucina, 2002](#)). Also, we have previously found that the results based on the Reid and

Table 3

LEH measurements (in mm) and age at LEH formation (in years) for each decile of the crown heights – mandible [calculated using tooth crown formation times from Table 4 in Reid and Dean (2006, p. 333)].

Crown height (%)	I1		I2		C	
	LEH measurement	Age	LEH measurement	Age	LEH measurement	Age
0–10	9.29–10.31	1.0–1.1	9.06–10.06	1.0–1.1	12.08–13.41	1.5–1.7
11–20	8.26–9.28	1.2–1.3	8.06–9.05	1.2–1.3	10.74–12.07	1.8–2.0
21–30	7.23–8.25	1.4–1.5	7.05–8.05	1.4–1.5	9.40–10.73	2.1–2.3
31–40	6.20–7.22	1.6–1.7	6.05–7.04	1.6–1.8	8.06–9.39	2.4–2.7
41–50	5.17–6.19	1.8–2.0	5.04–6.04	1.9–2.1	6.72–8.05	2.8–3.1
51–60	4.13–5.16	2.1–2.3	4.03–5.03	2.2–2.4	5.37–6.71	3.2–3.6
61–70	3.10–4.12	2.4–2.6	3.03–4.02	2.5–2.8	4.03–5.36	3.7–4.2
71–80	2.07–3.09	2.7–3.0	2.02–3.02	2.9–3.3	2.69–4.02	4.3–4.9
81–90	1.04–2.06	3.1–3.4	1.02–2.01	3.4–3.7	1.35–2.68	5.0–5.6
91–100	0–1.03	3.5–3.8	0–1.01	3.8–4.2	0–1.34	5.7–6.2

Dean's European crown heights and the ones based on our own sample's means were similar (Krenz-Niedbala and Łukasik, 2012). The LEHs were measured along a straight line, without correcting for surface curvature (cf. Reid and Dean, 2000, 2006). Guatelli-Steinberg et al. (2007) have provided such corrections for Inuit and Neandertal teeth, which seem to be of little use for our study, considering genetic distance between our sample and Inuit or Neandertal populations. Guatelli-Steinberg and co-workers' data were applied by Ritzman et al. (2008) and they found no differences between the results produced using measurements along a straight line and along the crown surface.

The hypoplastic defects were not matched between different tooth types, as it has been repeatedly observed that not all portions of enamel forming at the same time on different teeth will display hypoplasiae, due to, among others, greater susceptibility of polar teeth (more developmentally stable) to enamel developmental defects than nonpolar teeth (Goodman and Rose, 1990; King et al., 2005). Also, in order to compare all three methods at the same time, we had to include all defects observed in one individual, irrespective whether they represented the same chronological unit or not, because the Reid and Dean's deciles correspond to different time periods for different teeth (see Tables 2 and 3). It is possible to use an interpolation to produce the same estimated ages at defect formation, as Martin et al. (2008) have done, but this, according to the authors, introduces subjective bias into LEH formation estimates. Thus, although for the Goodman and Rose's and Goodman and Song's methods one can count only once the defects originating from the same stressful event (see Cucina and İşcan, 1997), this is not possible for the Reid and Dean's data-based method without interpolation. Thus, we have decided to include into the analysis all defects on all affected teeth from one individual.

The Reid and Dean's histological data (2000) at first included central and lateral incisors and canines, and then, molars (Reid and Dean, 2006). In a further study the data on premolar enamel formation times have been presented (Reid et al., 2008), but not for particular deciles, therefore they cannot be used for calculations of LEHs formation. Thus, the comparison of the three methods of LEH timing estimation was possible only for incisors and canines in this study.

All calculations were performed using the Microsoft Excel 2003 and Statistica 7.0 software.

Results

In the examined skeletal series linear enamel hypoplasia (Fig. 2) affected 40 individuals out of the total 77, which makes 51.9%. Although in the sample of Ślaboszewo the frequency of LEH is higher (57.1%) than in the Łekno population (45.7%), this difference is statistically insignificant. The hypoplastic defects tend to be more frequent in males than females (46.4% and 55.1%, respectively), but again, insignificantly.

There were 115 affected teeth out of 642 teeth included in the analysis, which makes 17.9%. In total, 148 hypoplastic lines were recorded. The highest frequency of LEH was found for canines (35.6%), then incisors (32.2%) and premolars (13.1%). Higher frequency was reported for mandibular (21.3%) than maxillary teeth (14.6%). This difference is statistically significant ($\chi^2 = 4.903$, $p = 0.0268$). The



Fig. 2. Linear enamel hypoplasia in individual 21/90, female, young adult, Staboszewo.

highest LEH frequency in maxillary dentition was found for central incisors (27.6%), and in mandibular dentition for lateral incisors (35.4%). The majority of LEH defects were reported for young adults (54.3%), then middle adults (50.0%), and the lowest frequency was found in old adults (40.0%). However, the difference between the first two age categories is not statistically significant, and old adults were not tested on the account on their low number.

In Fig. 3 there are numbers of defects by age categories estimated with the use of the regression equations proposed by Goodman and Rose and Goodman and Song. Both methods produced two peaks of defects, the conventional method at the age categories of 2.1–3.0 years and 4.1–4.5 years of age, and the corrected method at 2.1–3.5 and 4.1–5.0 years of age. The use of the corrected method extended the upper range of both peaks by 6 months toward later ages.

Reid and Dean's formation times of particular deciles and, in consequence, age ranges of LEH formation, differ between the tooth types (Reid and Dean, Tables 4 and 5, 2006:333–334), thus, in order to compare all three methods of LEH timing estimation it is necessary to consider separately each maxillary and mandibular tooth category. To make the comparison clearly visible and easily interpretable a diagram was constructed for each tooth type that shows at the same time age ranges of all the defects and of the peaks in LEH formation according to the three methods. Such charts allow for showing clearly to what extent the results provided by the three methods differ from one another

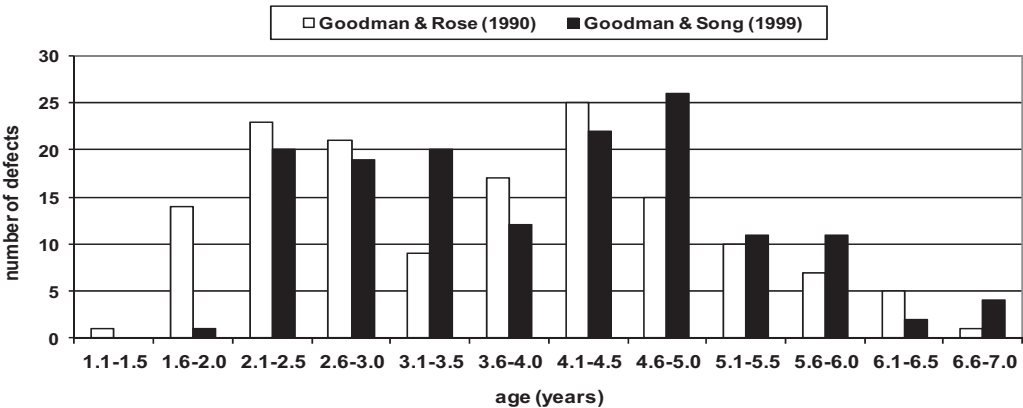


Fig. 3. Number of LEHs by age categories based on Goodman and Rose (1990) and Goodman and Song (1999) methods.

Table 4

LEH frequencies in various European osteological series; data arranged according to the LEH frequencies.

Site and dating	No. of individuals	Individuals with LEH (%)	Teeth with LEH (%)	Teeth examined	Data source
Eleutherna and Messene, Greece (6th–7th c. CE)	225	–	1.8	All	Bourbou (2003)
Monte d'Argento, Italy (12th c. CE)	60	–	25.1	–	Carrasco et al. (1992)
Composite, Croatia (6th–9th c. CE)	259	–	36.1	Anterior	Šlaus et al. (2002)
Composite, Austria (7th–9th c. CE)	360	–	49.0	All	Herold (2008)
Composite, Croatia (10th–14th c. CE)	155	–	49.2	Anterior	Šlaus et al. (2002)
Vilarnau, France (12th–14th c. CE)	–	–	63.0	–	Munoz et al. (2008)
Kołobrzeg, Poland (14th–18th c. CE)	190	22.4	–	Canines	Kozak and Krenz-Niedbala (2002)
Borovce, Slovakia (8th–12th c. CE)	302	27.2	7.8	All	Obertová (2005)
Gracanica and Dici, Serbia (14th–19th c. CE)	105	29.6	5.0	All	Djurić-Srejić (2001)
Nova Raca, Croatia (14th–18th c. CE)	75	35.6	33.4	Anterior	Šlaus (2000)
Milicz, Poland (12th–14th c. CE)	90	38.0	–	Anterior	Dąbrowski and Gronkiewicz (1996)
Cedynia, Poland (10th–13th c. CE)	324	40.1	20.7	Anterior	Piontek and Krenz-Niedbala (1999)
Dolní Vestonice, Czech Republic (11th–12th c. CE)	63	39.7	35.1	Anterior	Jarošová (2006)
Sypniewo, Poland (11th–13th c. CE)	135	41.5	–	All	Staniowski et al. (2004)
Nin, Croatia (12th–15th c. CE)	77	44.2	–	Anterior	Novak et al. (2012)
Dugopolje, Croatia (14th–16th c. CE)	195	49.7	–	Anterior	Novak and Šlaus (2007)
Staboszewo and Łekno, Poland (13th–18th c. CE)	77	51.9	17.9	All	This paper
Poznań, Poland (9th–13th c. CE)	54	60.0	–	Anterior	Betsinger (2007)
Composite eastern Adriatic, Croatia (7th–9th c. CE)	172	54.1	–	Anterior	Šlaus (2008)
Privlaka, Croatia (8th–9th c. CE)	153	67.5	–	All	Šlaus (1996)
Continental, Croatia (6th–9th c. CE)	103	74.8	–	Anterior	Šlaus (2008)
La Selviccola, Italy (7th c. CE)	44	75.0	23.1	All	Manzi et al. (1999)
Pada, Estonia (12th–13th c. CE)	117	77.8	–	Anterior	Limbo (2006)
Composite, Lithuania (13th–18th c. CE)	837	83.6	–	All	Palubeckaitė (2005)
Tirup, Denmark (12th–14th c. CE)	131	86.3	–	All	Palubeckaitė et al. (2002)
Composite, Lithuania (5th–7th c. CE)	189	87.3	–	All	Palubeckaitė (2005)
Composite, Lithuania (15th–18th c. CE)	66	93.5	–	All	Palubeckaitė et al. (2002)
Vilnius, Lithuania (16th–17th c. CE)	88	100.0	–	All	Palubeckaitė et al. (2002)
Vicenne, Italy (7th c. CE)	83	100.0	60.1	All	Belcastro et al. (2007)

Table 5
Age ranges of LEH peaks in selected samples.

Site and dating	Peak of LEH	Method	Data source
Lithuania, Denmark (12th–18th c. CE)	4 years	Massler et al. (1941)	Palubeckaitė et al. (2002)
Borovce, Slovakia (8th–12th c. CE)	2.5–3.0	Massler et al. (1941)	Obertová (2005)
Pada, Estonia (12th–13th c. CE)	2.5–3.0	Massler et al. (1941)	Limbo (2006)
Florence, Italy (19th c. CE)	2.5–3.0	Massler et al. (1941)	Moggi-Cecchi et al. (1994)
Sypniewo, Poland (11th–13th c. CE)	2.5–3.0	Massler et al. (1941)	Staniowski et al. (2008)
London, England (18th–19th c. CE)	2.0–4.0	Reid and Dean (2006)	King et al. (2005)
Vallerano, Italy (2nd–3rd c. CE)	3.0–3.4	Reid and Dean (2000)	Cucina et al. (2006)
Dolní Vestonice, Czech Republic (11th–12th c. CE)	3.5–4.0	Reid and Dean (2000)	Jarošová (2006)
Rogowo, Poland (2nd c. CE)	2.6–3.0	Goodman and Rose (1990)	Krenz-Niedbała and Kozłowski (2013)
Rogowo, Poland (2nd c. CE)	3.0–4.0	Goodman and Song (1999)	Krenz-Niedbała and Kozłowski (2013)
Łekno and Staboszewo, Poland (13th–18th c. CE)	2.1–2.5	Goodman and Rose (1990)	This paper
Łekno and Staboszewo, Poland (13th–18th c. CE)	3.1–3.5	Goodman and Song (1999)	This paper

despite different age categorization in each method. In Fig. 4 there is a comparison of the three methods of LEH timing estimation for maxillary incisors and canines, and in Fig. 5 for mandibular incisors and canines.

Different data were obtained for the upper central and lateral incisors in relation to age ranges and peaks of LEH. The decile method and Goodman and Song's method caused the shift of the results

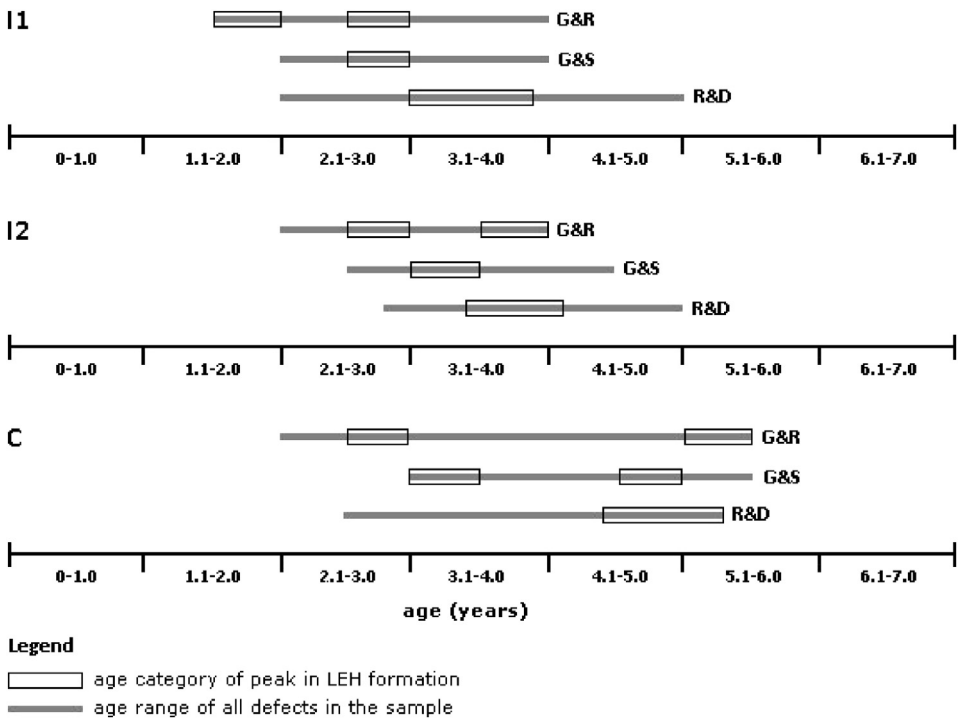


Fig. 4. Comparison of three methods of LEH timing estimation on maxillary teeth; G&R – Goodman and Rose (1990); G&S – Goodman and Song (1999); R&D – Reid and Dean (2000, 2006).

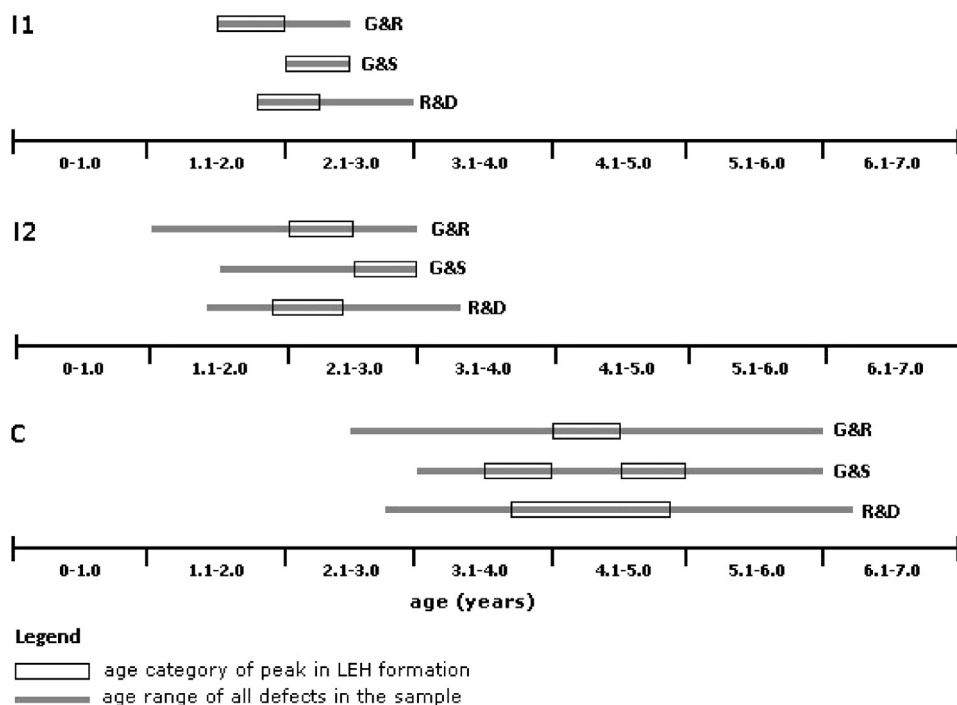


Fig. 5. Comparison of three methods of LEH timing estimation on mandibular teeth; G&R – Goodman and Rose (1990); G&S – Goodman and Song (1999); R&D – Reid and Dean (2000, 2006).

toward later ages than the Goodman and Rose's method. For both incisors the conventional method produced two peaks and the remaining methods one peak of LEH. The results for the upper canine proved even more varying, with the earliest LEH peak at 2.6–3.0 years of age and the latest at 5.1–5.5 years. Goodman and Rose's and Goodman and Song's methods provided two peaks, while the decile method revealed one peak.

Regardless of the method applied, the age ranges of LEH formation obtained for the mandibular incisors are similar and oscillating between ~1.6 and 3.0 years of age. For the first incisor the LEH peaks are also similar and fall on the second year of age. For the second incisors the data on LEH formation provided by the Goodman and Rose's and the decile methods are similar, with peaks falling on 2.1–2.5 years of age. The use of the Goodman and Song's method has shifted the peak toward later timing, 2.6–3.0 years of age. For the mandibular canine all methods have produced similar results, with peaks falling on ~4.1–5.0 years of age.

In general, the method using the Reid and Dean's data led to the shift of age at LEH formation toward later categories in comparison with the conventional and corrected methods. For the maxillary teeth the age range of LEH formation and peaks of LEH provided by the three methods differed more than for the mandibular teeth.

Discussion

For the two examined late medieval/early historical samples no statistically significant differences in LEH frequency have been observed, and this finding indicates comparable living conditions in both populations. The LEH frequencies in the examined populations against other late medieval and/or early historical human samples are presented in Table 4. The results of LEH prevalence obtained in

this study are slightly higher than in other Polish samples (except for Poznań), but they are still relatively low as compared to other European osteological series. This finding likely reflects a moderate level of environmental constraints and/or effective cultural buffering of stressors in the examined groups (Goodman, 1991) as compared to other European populations. However, large differences in prevalence of LEH and variation in LEH timing among archaeological samples (Table 5) possibly result only partly from differences in living conditions and stress levels, and partly from applied methods. In LEH studies various analytical approaches have been adopted in relation to the minimum number of observations, selection of teeth for the research, choice of the method for defect timing estimation, and the way of gathering and calculating the data. The process of dental development varies among human populations coming from different geographical areas and chronological periods, but only a few tooth developmental standards are being applied (Goodman and Song, 1999).

There are nearly as many methods of LEH timing estimation as there are researchers (for example Cucina et al., 1997, 2002; Goodman and Rose, 1990; Goodman and Song, 1999; Massler et al., 1941; Murray and Murray, 1989; Reid and Dean, 2000, 2006; Swärdstedt, 1966). Thus, potential differences between various methods should be borne in mind, because chronology reconstruction of tooth developmental defects is critical for the interpretation of stress experienced in childhood and the related phenomena (e.g. weaning). The main sources of variation in LEH timing estimation have been thoroughly discussed by Goodman and Song (1999), who emphasized the role of hidden cuspal enamel in producing biased results.

The analysis performed in this study revealed considerable differences in age at enamel defect formation introduced by the use of three different methods: Goodman and Rose (1990), Goodman and Song (1999), and the one based on Reid and Dean's data (2006), related to the whole range of defect chronology, range of peaks, number of peaks and age category of peaks. The use of the decile method led to the shift of the timing toward older age than obtained with both conventional and corrected methods, in accordance with the results of Ritzman et al. (2008), Krenz-Niedbała and Kozłowski (2013), and, partly, Martin and co-workers (2008). Moreover, the study of Goodman and Song (1999) has shown that the greatest differences in the estimation of age at LEH formation by the regression equations without and with the correction for hidden cuspal enamel occurred for the earliest periods of enamel development. This is true also for the decile method, since different results it provides come from including the time of cuspal enamel development. The ages at LEH defects formation for later deciles are much more similar to the results obtained with the use of both conventional and corrected methods. This suggests that the differences related to the defect timing would be much more apparent in the samples, where the lines were formed mainly at the earliest stages of tooth development (Ritzman et al., 2008).

The chronological distribution of LEH defects provides information on the periods when growing children were particularly exposed to adverse environmental factors. Although the defects formed after the first year of age begin to be visible on the crown surface (Reid and Dean, 2006), the earliest stress episodes (according to the methods correcting for buried enamel) occurred in the samples examined in this study only after the second year of life. This likely shows the protective effect of breastfeeding, since many of human milk's ingredients, such as immune factors, anti-inflammatory agents, enzymes, and viable cells enhance infant's immunological system (Hanson, 2007).

The results of numerous studies on LEH chronology based on the Massler and co-workers' mineralization diagram (1941) carried out on both European and American populations show that the peak of stress falls on the age category 2–3 years. The studies using the model of Reid and Dean (2000, 2006), note the peak at LEH slightly later – between 2 and 4 years of age (Table 5). The use of more precise method certainly ensures more proper interpretation of the obtained results, but, anyway, inferences about weaning based on a peak in LEH formation should be treated with caution when the information from additional sources, for example historical archives, is not available (Goodman and Song, 1999).

In this study the analysis of chronological distribution of enamel hypoplasiae based on the Massler and co-workers' standard revealed two peaks of stress. According to the Goodman and Rose's method they fall on 2.1–2.5 and 4.1–4.5 years of age, and according to the Goodman and Song's method 2.1–3.5 and 4.5–5.0 years of age. The use of the corrected method shifted the peaks toward higher age categories. The early peak of LEH is usually regarded as being related to the consequences of weaning the

child, associated with many adverse factors, such as loss of nutrients from the mother's milk, decrease in the immunological resistance, poor quality of solid foods, poor sanitary conditions, increasing mobility of the child, and exposure to environmental hazards (Huss-Ashmore et al., 1982; Krenz-Niedbała, 2000; Lanphear, 1990; Larsen, 1987; Moggi-Cecchi et al., 1994; Ritzman et al., 2008). Weaning might be abrupt, but usually is a gradual process of introducing complementary foods to an infant's diet. The protective effect of breastfeeding is still provided when the child receives most of their nutrition from other sources than human milk (World Health Organization, 2004). Thus, the crucial moment for the child's health is complete discontinuation of breast milk feeding.

Average age of weaning of 2–3 years in medieval Poland has been documented by historical data (Delimata, 2004; Żołędź-Strzelczyk, 2002). Medical, philosophical and pedagogical writings of Middle Ages recommended prolonged breastfeeding, regarded as most suitable and healthy nutrition for the infant and also beneficial for the mother, being even a symbol of maternal love (Delimata, 2004). It was advised to wean the child gradually, in time and manner that adhere to the child's needs (Żołędź-Strzelczyk, 2002). Breastfeeding cessation at 2–3 years corresponds to the results of the present study, concurring with most of the data calculated according to the Reid and Dean's crown formation times (see Figs. 4 and 5). The second peak of LEH (in so-called post-weaning stage), as revealed using the Massler and co-workers' standard, could be interpreted in terms of the family model. In the examined sample the difference between the first and the second peak was 2 years for the conventional method and 2.5 year for the corrected method. In the Middle Ages there was high fertility, and the distance between the subsequent deliveries was just about 2.5 years (Żołędź-Strzelczyk, 2002). Arrival of a new baby in the family might have caused a decrease in the care for the former one, and together with increasing exploratory activities of the child at the age of 4–5 years might have led to greater exposure of older children to a variety of stressors. However, there are also other potential factors that may account for the second peak, such as increased incidence of infectious diseases in early childhood, and participation of the child in some minor household chores (Lewis, 2007), which might have exposed them to health hazards such as indoor air pollution (Barnes et al., 2005). Most likely, all of those disturbing factors acted in combination to have caused enamel developmental defects.

However, the decile method has shown a different picture of the childhood stress experience than both methods based on Massler and co-workers' standard. Defect distribution is unimodal and the peaks are extended, as compared with Goodman and Rose's and Goodman and Song's methods. It seems as if the starting point for the increased stress was weaning and then stressful events (some of them listed in the previous paragraph) continued to negatively affect the child's growth and development until about 5 years of age. This view better fits what we know about the everyday life of children in the Middle Ages, in reference e.g. to limited diet, poor sanitary conditions, and low parental control (Delimata, 2004; Żołędź-Strzelczyk, 2002).

The smallest differences in the results produced by the methods of LEH timing estimation, based either on Massler and co-workers' or Reid and Dean's data, have been found for mandibular teeth, especially canine. This is, at the same time, the tooth category most recommended for LEH studies (Buikstra and Ubelaker, 1994) as being particularly susceptible to stressors causing developmental enamel defects (Goodman and Rose, 1990). Thus, when time does not permit recording defects on all teeth available, mandibular canines ought to be chosen for the study (Buikstra and Ubelaker, 1994).

In summary, it should be emphasized that three methods of LEH timing estimation applied in this study provided different results related to the entire range of defect chronology, range of peaks, number of peaks and age category of peaks. The Reid and Dean's crown formation times based on tooth histology constitute more precise and accurate standard than those previously used. This proved true for the samples studied, considering the greatest correspondence between the results obtained and the data recovered from historical sources.

Conclusions

The analysis carried out in this study allows to infer that the environment of the examined human medieval populations favored the correct course of developmental processes, since the frequency of linear enamel hypoplasia seems to be relatively low compared to other European odontological series. The choice of the method of LEH timing estimation proved to considerably influence later

interpretation of the results. The use of the method based on Reid and Dean's data (2000, 2006) led to the shift in LEH timing toward the age categories higher than those obtained using both the conventional and corrected methods. Greater differences related to the age range of the defects and peaks in LEH formation provided by different methods were found for maxillary than mandibular teeth. Taking into account more methodologically sound approach of Reid and Dean it seems advisable to use their standard when calculating ages of enamel hypoplasia formation.

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